

ODE

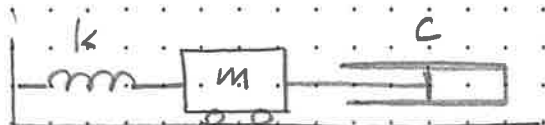
$$y'' + p(t)y' + q(t)y = f(t)$$

WITH DAMPING

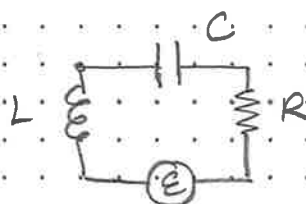
$$m \cdot y'' + C y' + k y = f(t)$$

$$L_2' + R_2 + \frac{q}{C} = \mathcal{E}(t)$$

$$\Rightarrow L \ddot{z} + R \dot{z} + \frac{z}{\lambda} = \varepsilon(t)$$



[SPRING-MASS-DASHPOT.]
[SYSTEM]



[R-L-C CIRCUIT]

$$\begin{bmatrix} y(t) \\ = y_p + y_h \end{bmatrix}$$

Particular solution

Homogeneous Solution . .

Like for
→ a dry cell,
we have
a homogenous
equation

$$[\dot{z}(t)] = \dot{z}_p + \dot{z}_h$$

Particular Solution

1. Ito mo -
general
solution

Both these applications can have a special case with constant coefficients

Aside:

$f(t) \neq 0$	Forced system Non-homogeneous system
$f(t) = 0$	Passive system Homogeneous system

$$y'' + Cy' + By = f(t)$$

$\{ \text{constants} \}$

COMPLETE SOLUTION

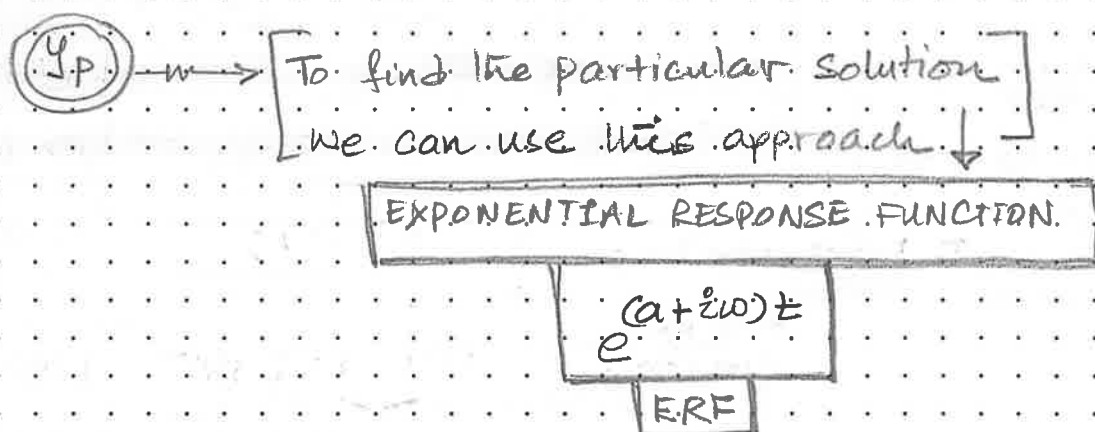
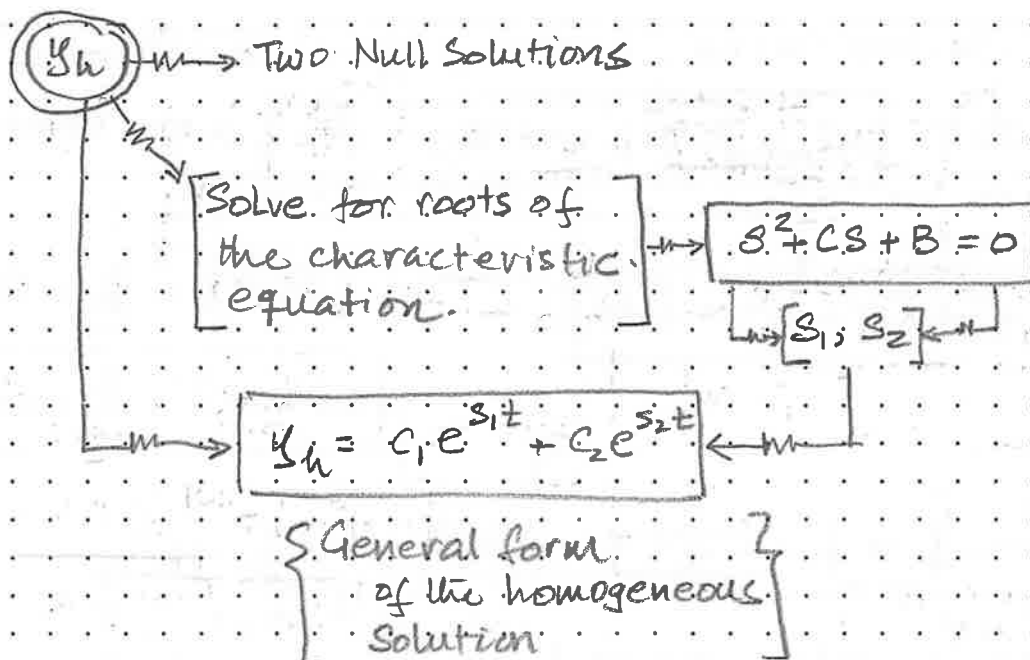
$$-Y(t) \cdot$$

$$y(t) = y_p + C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

where y_p : Particular solution

y_h : Homogeneous solution ($C_1 e^{s_1 t} + C_2 e^{s_2 t}$)
 s_1, s_2 : Roots of the characteristic equation

S_1, S_2 : Roots of the characteristic equation



We need to develop some basic understanding about the forcing term, $f(t)$ can be expressed as the exponential response function, $e^{(a+i\omega)t}$.

Some examples of $f(t)$:

e^{at}

$a < 0$

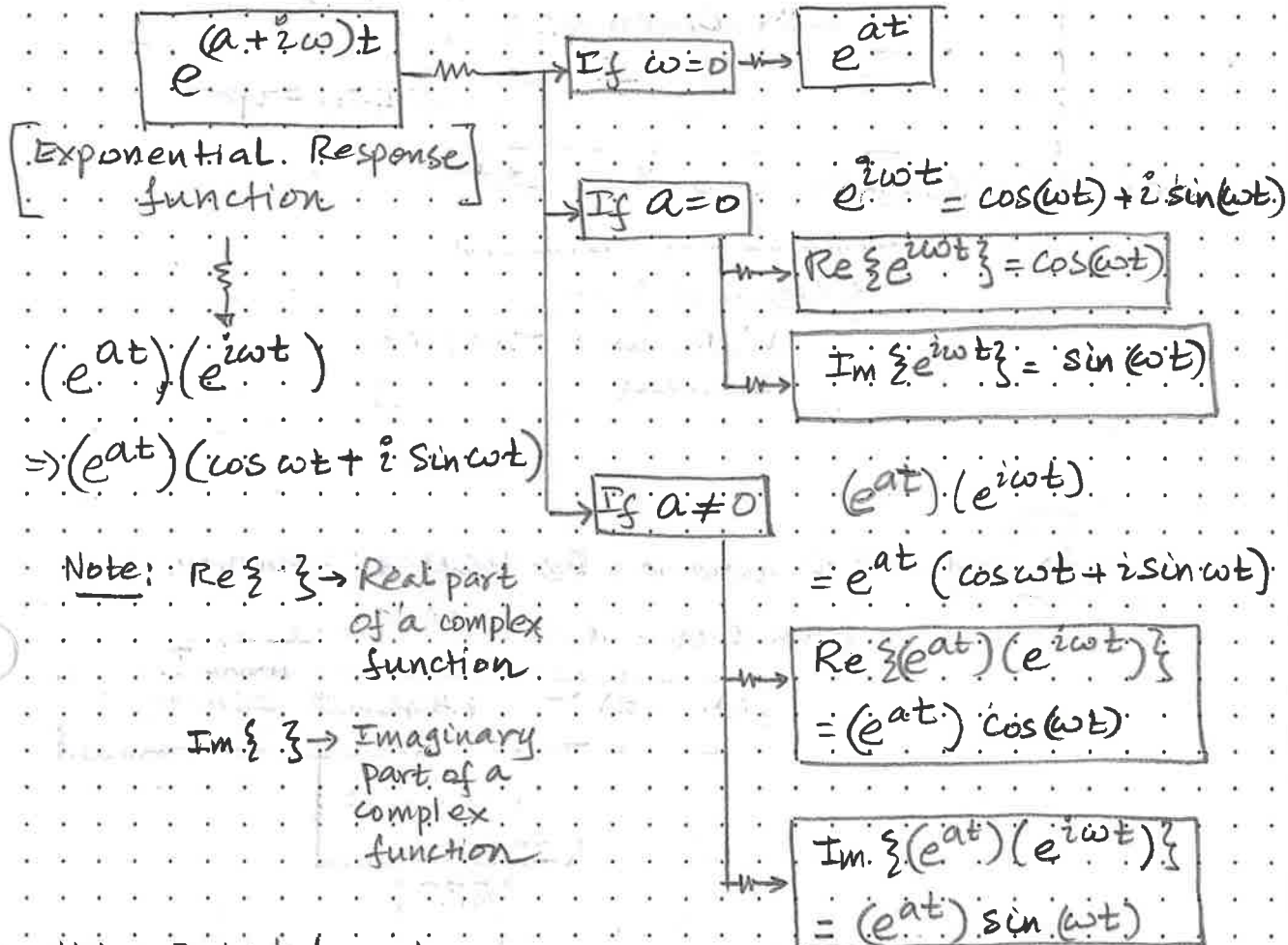
typically a decaying function

$\sin(\omega t)$
 $\cos(\omega t)$

$e^{at} \sin(\omega t)$
 $e^{at} \cos(\omega t)$

We can express some typical (forcing) functions with the ERF:

Typical Forcing functions



Note: Euler's formula

$$e^{i\omega t} = \cos(\omega t) + i \sin(\omega t)$$

To find the particular solution, y_p , we are going to use the "special" function. We have the ERF, $e^{(a+i\omega)t}$.



Your professor is trying to make this easy to get to.



This approach is amazingly easy to implement; whenever we can write $f(t)$ as an ERF,

Here's some background in 5 points, to get you started.

First, write down the 2nd order, linear, non-homogeneous ODE in the following form:

$$y'' + Ay' + By = f(t)$$

where A, B are constants.

Second, write down the ODE as polynomial operator acting on $y(t)$.

$$(D^2 + AD + B)y = f(t)$$

$p(D)$

Polynomial operator



Linear operator

D : retains the idea of a derivative

$$D = \frac{d}{dt}, D^2 = \frac{d^2}{dt^2}, D^3 = \frac{d^3}{dt^3}, \dots$$

Aside:

Recall:

The characteristic equation for the homogeneous (equivalent) ODE is

$$s^2 + As + B = 0$$

$p(s)$

$p(s)$ is same as $p(D)$!

Note: Currently, $p(D)$ is a 2nd degree, linear, polynomial operator, representing the second order ODE. $p(D)$ can be of a higher degree, representing a higher order, linear, ODE.

Third, the particular solution, y_p , should, if correct, satisfy the ODE.

KVB

$$y_p'' + Ay_p' + By_p = e^{\alpha t} \quad \text{where } e^{\alpha t} = e^{(a+iz\omega)t}$$

$$\Rightarrow (D^2 + AD + B)y_p = e^{\alpha t} \quad \text{or } p(D)y_p = e^{\alpha t}$$

Replacing $f(t)$ with $e^{\alpha t}$; "Complexifying the ODE"

Fourth, write the particular solution in the following form:

$$y_p = \frac{e^{\alpha t}}{p(\alpha)} \quad \text{where } e^{\alpha t} = e^{(a+iz\omega)t}$$

Aside: Basic formula - Substitution Rule

$$p(D)e^{\alpha t} = p(\alpha)e^{\alpha t} \quad \text{where } e^{\alpha t} = e^{(a+iz\omega)t}$$

(i) Apply $p(D)$ to any complex exponential function $e^{\alpha t}$;

(ii) You substitute α , for D , and that's it!

Finally, let's check why $y_p = \frac{e^{\alpha t}}{p(\alpha)}$ works as the particular solution.

Recall that we expressed $f(t)$ as some $e^{\alpha t}$ where $\alpha = a + iz\omega$.

$$\therefore p(D)y_p = e^{\alpha t} \quad \{\text{from the "third" point}\}$$

$$\Rightarrow p(D) \frac{e^{\alpha t}}{p(\alpha)} = e^{\alpha t} \quad \{\text{using the substitution rule}\}$$

$$p(D)e^{\alpha t} = p(\alpha)e^{\alpha t}$$

$$\text{L.H.S: } p(D) \frac{e^{\alpha t}}{p(\alpha)} = \frac{p(\alpha)e^{\alpha t}}{p(\alpha)} = e^{\alpha t} \Rightarrow \text{L.H.S} = \text{R.H.S}$$



Let's pull everything together
in some examples

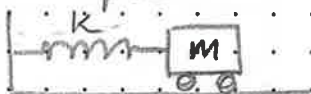
KVB

EXAMPLE-1

Solve $\ddot{y} + 10y = \cos \omega t$

for $\omega^2 \neq 10$

Solution: This is problem of an undamped, forced oscillator.



We need: y_p & y_h for the complete solution, $y(t)$.

$$[y(t) = y_p + y_h]$$

(I) Homogeneous solution.

① Characteristic equation: $p(s) = 0$

$$s^2 + 10 = 0$$

$$\Rightarrow s = \pm \sqrt{10} i \Rightarrow \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} = \begin{bmatrix} +i\sqrt{10} \\ -i\sqrt{10} \end{bmatrix}$$

$$\text{② } y_h = C_1 e^{s_1 t} + C_2 e^{s_2 t}$$

$$\Rightarrow y_h = C_1 e^{i\sqrt{10}t} + C_2 e^{-i\sqrt{10}t}$$

Using Euler's formula

$$y_h = C_1 (\cos \sqrt{10}t + i \sin \sqrt{10}t) + C_2 (\cos \sqrt{10}t - i \sin \sqrt{10}t)$$

$$\Rightarrow y_h = (C_1 + C_2) \cos \sqrt{10}t + (C_1 - C_2) i \sin \sqrt{10}t$$

$$\Rightarrow y_h = C_1 \cos(\sqrt{10}t) + C_2 \sin(\sqrt{10}t)$$

General solution:

$$[y(t) = y_h + y_p]$$

$$y(t) = C_1 \cos(\sqrt{10}t) + C_2 \sin(\sqrt{10}t) + \left(\frac{1}{10 - \omega^2}\right) \cos(\omega t)$$

(II) Particular solution

① Express forcing term as $e^{i\omega t}$

$$\cos \omega t = \text{Re} \{ e^{i\omega t} \}$$

② Complexify the ODE:

$$\ddot{y} + 10y = \cos \omega t \Rightarrow \ddot{z} + 10z = e^{i\omega t}$$

$$\& y_p = \text{Re} \{ z_p \} \text{ solves } \ddot{y} + 10y = \cos \omega t$$

③ Write down the particular solution: $z_p = \frac{e^{i\omega t}}{p(i\omega)}$

$$\Rightarrow z_p = \frac{e^{i\omega t}}{p(i\omega)} \quad \because p(s) = s^2 + 10 \Rightarrow p(i\omega) = (i\omega)^2 + 10$$

$$\Rightarrow z_p = \frac{e^{i\omega t}}{10 - \omega^2}$$

④ From step ③ $y_p = \text{Re} \{ z_p \}$:

$$y_p = \text{Re} \left\{ \left(\frac{1}{10 - \omega^2} \right) e^{i\omega t} \right\} \\ = \text{Re} \left\{ \left(\frac{1}{10 - \omega^2} \right) [\cos(\omega t) + i \sin(\omega t)] \right\}$$

$$y_p = \left(\frac{1}{10 - \omega^2} \right) \cos(\omega t) \quad \text{where } \omega^2 \neq 10$$

where $\omega^2 \neq 10$

EXAMPLE - 2 Use ERF to find the particular

solution of the ODE: $\ddot{y} + 2\dot{y} + 4y = \cos(3t)$

Solution:

(I) Express forcing term as e^{xt} where $x = a + i\omega$:

$$\text{Let } \cos(3t) = \operatorname{Re}\{e^{i3t}\}$$

(II) Complexify the ODE:

$$\ddot{y} + 2\dot{y} + 4y = \cos(3t) \Rightarrow \ddot{z} + 2\dot{z} + 4z = e^{i\omega t}$$

$$\& y_p = \operatorname{Re}\{z_p\}$$

$$\text{solves } \ddot{y} + 2\dot{y} + 4y = \cos(3t)$$

(III) Write the particular solution: $[z_p = \frac{e^{xt}}{p(x)}]$

$$z_p = \frac{e^{i3t}}{p(3i)}$$

$$= \frac{e^{i3t}}{-5+6i}$$

$$\because p(s) = s^2 + 2s + 4$$

$$\Rightarrow p(3i) = (3i)^2 + 2(3i) + 4$$

$$= -9 + 6i + 4$$

$$= -5 + 6i$$

(IV) From step (II) $y_p = \operatorname{Re}\{z_p\}$:

$$y_p = \operatorname{Re}\left\{\frac{e^{i3t}}{-5+6i}\right\} = \frac{\cos(3t) + i\sin(3t)}{-5+6i}$$

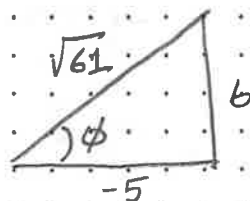
$$y_p = \operatorname{Re}\left\{\frac{(-5-6i)}{(-5-6i)} \left[\frac{\cos(3t) + i\sin(3t)}{-5+6i}\right]\right\} \leftarrow \text{Multiply \& dividing by the complex conjugate of } -5+6i$$

$$= \operatorname{Re}\left\{\frac{(-5-6i)}{25+36} [\cos(3t) + i\sin(3t)]\right\}$$

$$y_p = \left(\frac{1}{61}\right) [-5\cos(3t) + 6\sin(3t)]$$

$$\Rightarrow y_p = \left(\frac{\sqrt{61}}{61}\right) \cos(3t - \phi)$$

$$\Rightarrow y_p = \left(\frac{1}{\sqrt{61}}\right) \cos(3t - \phi) \text{ where } \phi = \tan^{-1}\left(\frac{6}{5}\right)$$



EXAMPLE-3

Find the particular solution of

the ODE: $y'' - y' + 2y = 10e^{-t} \sin t$

Solution: This problem is of decaying oscillator since there is a decay exponential e^{-t} in the forcing term

(I) Express the forcing term as $e^{\kappa t}$ where $\kappa = a + i\omega$:

Let $10e^{-t} \sin t = \text{Im} \{ 10e^{(-1+i)t} \}$

Aside:

$$\begin{aligned} (10)e^{(-1+i)t} &= (10)(e^{-t})(e^{it}) \\ &= (10)(e^{-t})(\cos t + i \sin t) \end{aligned}$$

Imaginary part of this is $10e^{-t} \sin t$

(II) Complexify the ODE:

$$y'' - y' + 2y = 10e^{-t} \sin t$$

$$\Rightarrow z'' - z' + 2z = 10e^{(-1+i)t}$$

& $y_p = \text{Im} \{ z_p \}$ solves $y'' - y' + 2y = 10e^{-t} \sin t$

(III) Write the particular solution:

$$z_p = \frac{e^{\kappa t}}{p(\kappa)}$$

$$z_p = \frac{10e^{(-1+i)t}}{p(-1+i)}$$

$$= \frac{10e^{(-1+i)t}}{3-3i}$$

$$= \left(\frac{10}{3}\right) \left[\frac{e^{(-1+i)t}}{1-i} \right]$$

$$p(s) = s^2 - s + 2$$

$$p(-1+i) = (-1+i)^2 - (-1+i) + 2$$

$$= (-1)^2 + 2i(-1) + i^2 + 1 - i + 2$$

$$= 1 - 2i + i^2 + 1 - i + 2$$

$$\Rightarrow p(-1+i) = 3-3i$$

$$= \left(\frac{10}{3}\right) \left(\frac{1+i}{1-i}\right) \left[\frac{e^{(-1+i)t}}{1-i} \right]$$

→ Multiplying the numerator and denominator by the complex conjugate of $(1-i)$

$$= \left(\frac{10}{3}\right) \left(\frac{1}{2}\right) (e^{-t})(1+i)e^{it}$$

$$= \left(\frac{10e^{-t}}{6}\right) (1+i)(\cos t + i \sin t)$$

④ From Step ② $y_p = \text{Im} \{ z_p \}$:

$$y_p = \text{Im} \left\{ \left(\frac{\sqrt{5}}{3} e^{-t} \right) (1 + i) (\cos t + i \sin t) \right\}$$

$$\Rightarrow y_p = \left(\frac{\sqrt{5}}{3} \right) (e^{-t}) (\cos t + \sin t)$$

$$\Rightarrow y_p = \left(\frac{\sqrt{5}}{3} \right) (e^{-t}) \sqrt{2} \cos(t - \phi)$$

$$\text{where } \phi = \frac{\pi}{4}$$

